CAPSTONE PROJECT

Student Performance Analysis Dashboard using Power BI

PRESENTED BY

STUDENT NAME: SANDEEP RATHOR D

COLLEGE NAME: VEMANA INSTITUTE OF TECHNOLOGY

DEPARTMENT: INFORMATION SCIENCE AND ENGINEERING

EMAIL ID: rathorsandeep4507@gmail.com

OUTLINE:

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PROBLEM STATEMENT:

Educational institutions need an efficient way to analyze student performance. It is difficult to track marks, identify top performers, and compare results across subjects. Manual analysis is time-consuming and less effective.

PROPOSED SOLUTION:

- The proposed system aims to analyze student performance using an interactive dashboard developed in Power BI. The system provides a clear and efficient way to visualize academic data and generate meaningful insights. It is designed to help users understand student performance across different subjects, categories, and criteria.

  Data Collection:
  The dataset is prepared in Microsoft Excel, containing student details such as name, gender, class, and subject-wise marks. This structured dataset acts as the input for analysis.

- **Data Processing:**
  The collected data is imported into Power BI, where it is cleaned and transformed. Calculated columns such as Total Marks, Percentage, and Result (Pass/Fail) are created using DAX formulas to enhance analysis.

- **Data Visualization:**
  The system uses various visual elements such as KPI cards, bar charts, column charts, and pie charts to represent data. Key insights include total number of students, average percentage, highest marks, subject-wise performance, and pass vs fail distribution.

- **Interactive Dashboard:**
  Filters (slicers) are added for Gender, Class, and Result, allowing users to interact with the dashboard dynamically. This enables users to view customized insights based on selected criteria.

- **Output & Insights:**
  The final dashboard provides a comprehensive overview of student performance, helping in identifying trends, comparing results, and making informed academic decisions.

SYSTEM APPROACH:

- **Technology Used:**
- Power BI Desktop
- Excel Dataset
- **System Requirements:**
- Windows OS
- Power BI installed
- Dataset for analysis

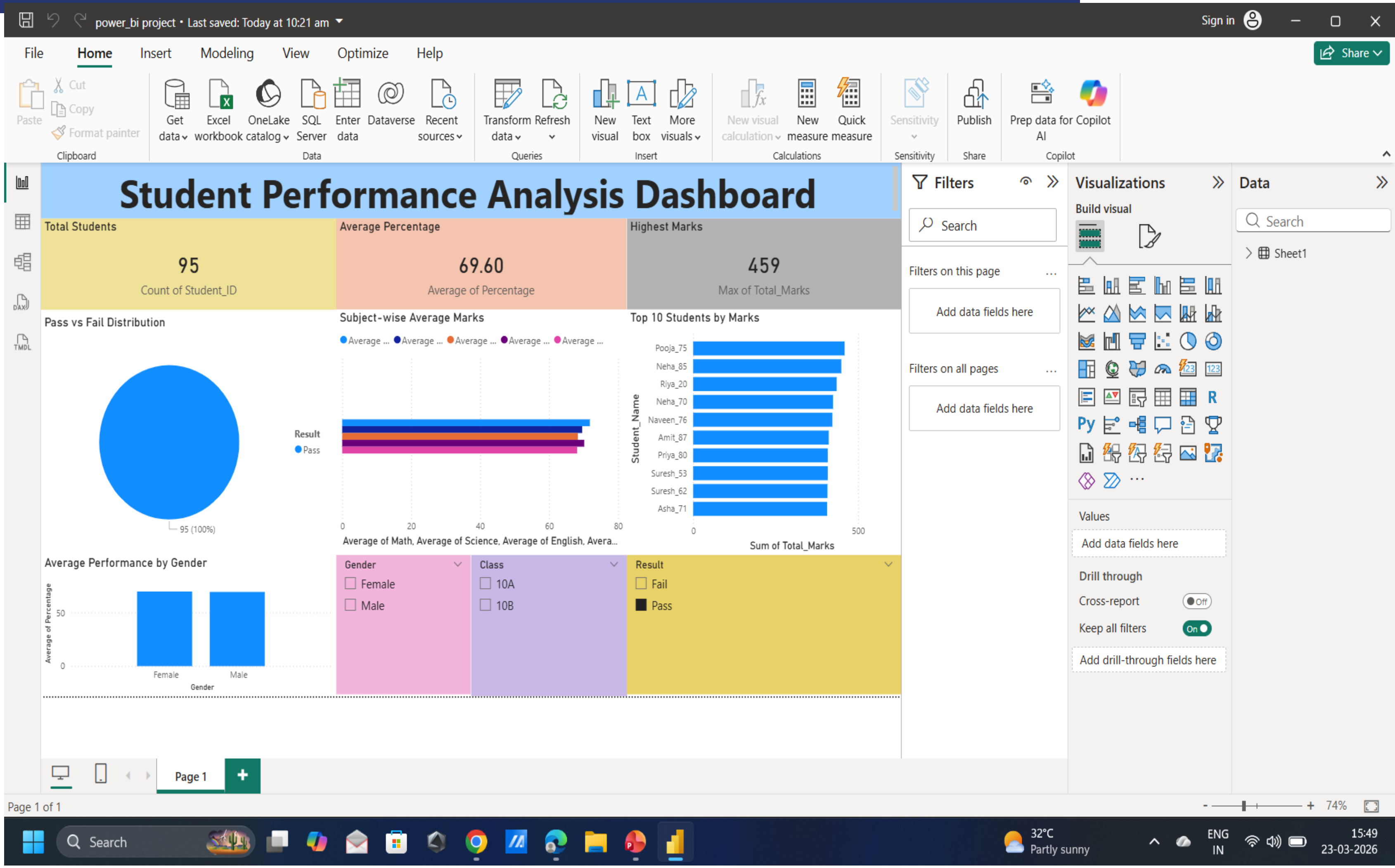
ALGORITHM & DEPLOYMENT:

- The project uses data analysis techniques in Power BI. Data is imported from Excel and processed using DAX formulas like Total Marks, Percentage, and Result. Visualizations are created using charts and slicers, and the dashboard is exported for reporting.

Microsoft Elevate

RESULT:

- The dashboard displays:
- Total students
- Average percentage
- Highest marks
- Pass vs Fail distribution
- Subject-wise performance
- Top 10 students
- Gender-based analysis



CONCLUSION:

This project demonstrates how Power BI can be used effectively for student performance analysis. It provides clear insights and helps in better academic decision-making through interactive visualizations.

FUTURE SCOPE:

Future improvements can include real-time data integration, advanced analytics, predictive modeling, and expansion to multiple classes or institutions.

REFERENCES:

- Microsoft Power BI Documentation
  Microsoft Learn
  Online tutorials and datasets
  GitHub Link: https://github.com/Sandeep-Rathor-D/student-performance-powerbi

Thank You